19 By completing the square, find the coordinates of the turning point of the curve with equation $y = x^2 + 10x + 18$
You must show all your working.

×	x	5
x	x^2	$5x$
5	$5x$	25

$(x+5)^2 = x^2 + 10x + 25$

1 mark
Half coefficient10

$y = (x+5)^2 - 25 + 18 = (x+5)^2 - 7.$

1 mark
Subtract7

Minimum when $x+5=0$: turning point$(-5,-7)$.

1 mark
$(-5,-7)$

(..... , ...)

(Total for Question 19 is 3 marks)